

Hypothesis Testing

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- What is an hypothesis?

An assumption, an idea, something proposed for the sake of an argument; an educated guess (not backed by data)

- The ^{contrast} opposite of hypothesis, then, is theory

↳ principle to explain things
backed by data

- Simple hypothesis testing example:

prob: We have a team of 4 members in a course project

Every week we meet once and present project status to Supervisor

Only one person needs to present ppt to Supervisor

Team decides to have a bowl, with names on chit, whose name comes he/she will present that week.

after 3 weeks: Bob never got to present

Note! things are assumed to be truly randomly.

Assume: Assume truly random (H₀: Null hypothesis)

$$P(\text{Bob not picked for a week}) = \frac{3}{4}$$

$$P(\text{Bob not picked 3 weeks in a row}) = \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} = \left(\frac{3}{4}\right)^3 = \frac{27}{64}$$

Let's say Bob did not present the entire sem!! = 0.42

$$P(\text{Bob not picked 12 weeks in a row})$$

$$= \left(\frac{3}{4}\right)^{12} = 0.03167$$

Hmm! Something is fishy. ⇐

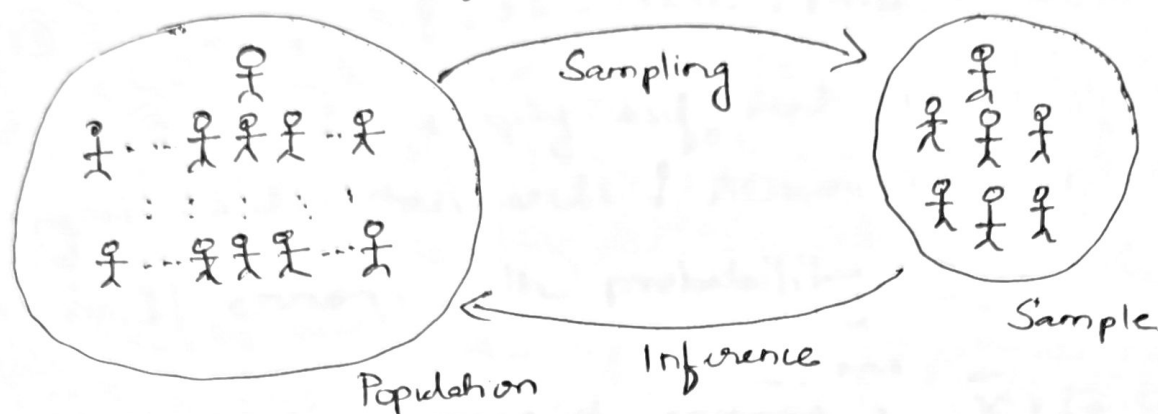
$$\approx \underline{0.032} (\approx 3\%)$$

Statisticians usually have thresholds,

if the prob. of something happening by chance is BELOW THRESHOLD
then reject the hypothesis (H₀)

Usage of hypothesis testing?

- Used to determine whether there is enough evidence in a sample of data to infer that a certain condition is true for the entire population. (this is part of inferential statistics)



Method to test an assumption about a parameter of a population based on a sample

- Steps:

1. Create a statement, a hypothesis, that we want to test
2. Population and Sample
3. Hypothesis test (t-test, ANOVA, ...)
4. Set up null hypothesis (usually pessimistic)
- 5a. Assume null hypothesis is true
- 5b. Calculate how much the sample deviates from Null hypothesis
- 5c. the p-value indicates how statistically significant our calculations on the sample are (A low p value indicates that the calculation is not a fluke, and is significant) Generally $p < 0.05$
- 5d. if $p < \text{threshold}$, means unlikely to get a calculation by chance
 $p > \text{ " }$, likely to get the calculation (so chances are high)

Errors:

Decision \ H_0	H_0	
	True	False
Accepted	Correct	Type II error
Rejected	Type I error	Correct